University of Lancashire

Coursework Cover Sheet

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Department and assessment information:

School Name: Engineering and Computing

Assessment title: Design Documentation

Course Title: BSc(Hons) Software Engineering

Module Title: Cross platform Development

Module Code: CO2404

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Year of Study: 2025

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I have applied the learning outcomes for this module ☐

I have checked for Academic Integrity via Turn-it-in ☐

I have followed the guidance in the Assessment Brief and have not used AI to boost my grade unfairly. ☐

I have used references in accordance with instructions in the Assessment Brief ☐

I have proofread my work for spelling, grammar and punctuation. ☐

I have checked that the word count/size of this submissions accords with the guidance provided in the Assessment Brief.☐

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Do you need to apply for mitigating circumstances for this assignment Please select Yes / No

Please refer to the Mitigating Circumstances Policy

Questions you may wish to consider:

1. Have I allowed sufficient time to prepare this assessment?
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3. What grade am I expecting?

Design Documentation

CosmicX

Mirco Fernando – G21319526

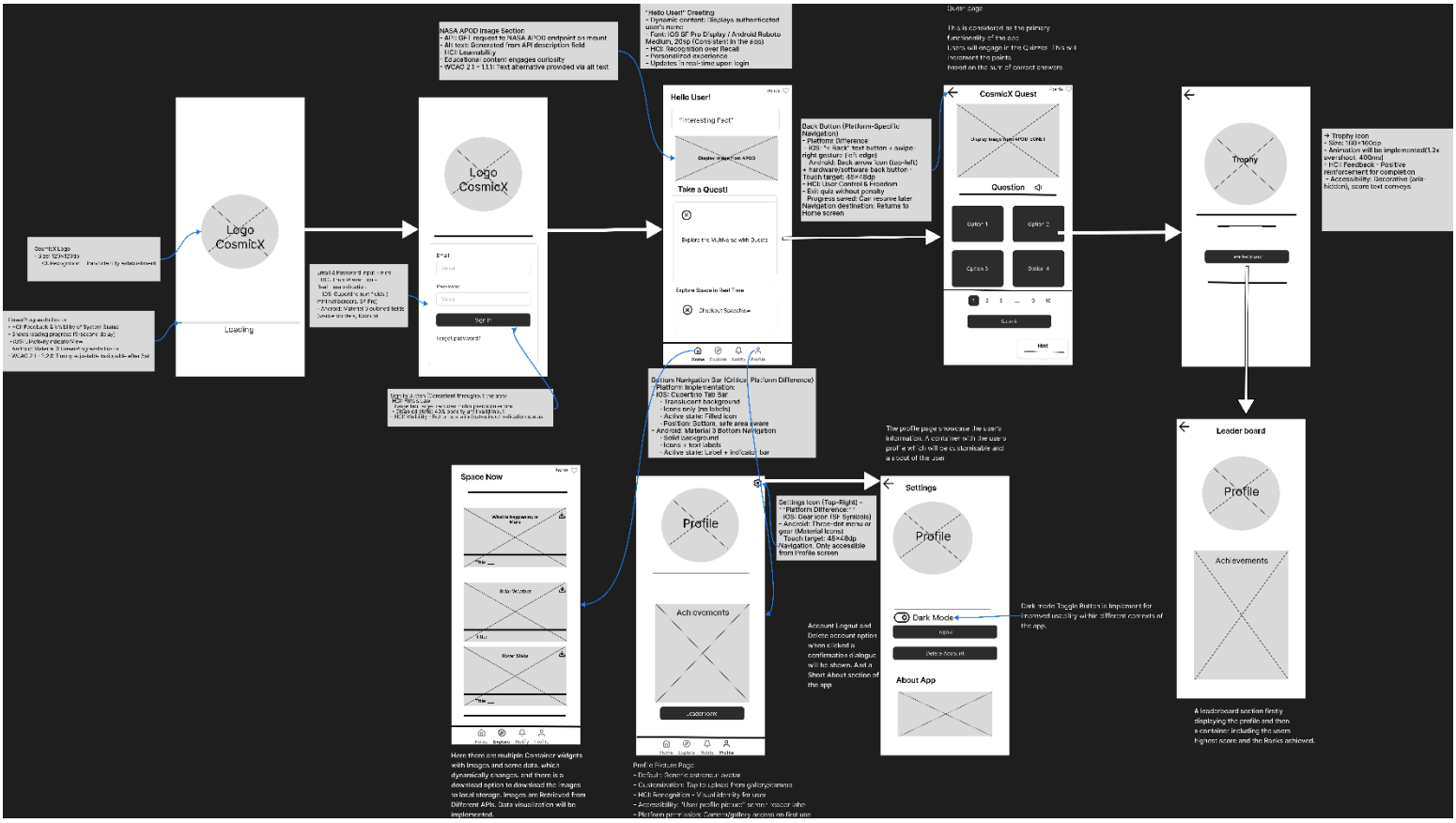
Table of Contents

1. Introduction 2
2. Wire frames 3
2.1 Additional Explanations 4
2.1.1 Loading Screen 4
2.1.2 Login Page 5
2.1.2 Home Screen 6
2.1.3 Quest Screen 7
2.1.4 Space Now explore 8
2.1.5 Points 9
3. HCI Principles Justification 10
3.1 Usability 10
3.2 Accessibility 12
3.3 Cross Platform Consistency 13
4. Conclusion 13
5. References 14

1. Introduction

As my solution I have chosen to develop an application aimed at engaging children, students and young space enthusiasts who have a keen interest in the space world. “CosmicX” is the name of this application. Recent studies have mentioned that kids and young people are mostly fascinated by space studies because it’s filled with curiosity and mysteries. (Communication Astrophilia, 2024). CosmicX lets users interact with quizzes, live NASA data, and collectable badges to nurture the curiosity to explore, learn, and entertain. The design emphasizes usability, accessibility, and consistency across Android and IOS devices, that reflects Human Computer Interaction (HCI) principles. Adhering to HCI principles are a core aspect of software design, ensuring usability, accessibility and an efficient user experience. “It ensures that technology is usable, accessible and efficient leading to better user experience.” (Sakoane, 2023). This design focuses to combine learning and entertainment by making exploration of space interactive and visually appealing.

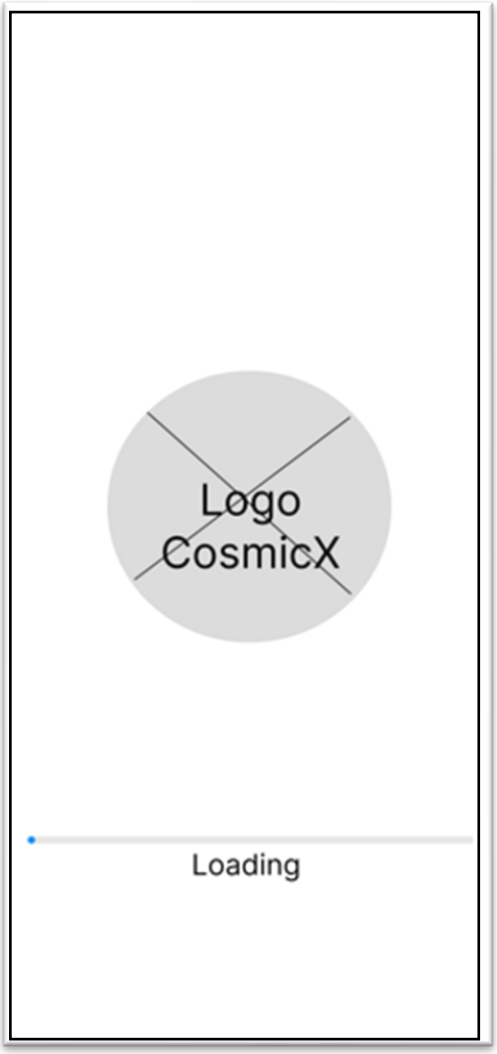
2. Wire frames



LINK:
https://drive.google.com/file/d/1CMpEFvQp2BpWInnYwoY6SF_u7yiqksin/view?usp=sharing

2.1 Additional Explanations

2.1.1 Loading Screen



The entry point of the application is the Loading Screen, this establishes a brand identity while loading the app environment. A fixed 5 second delay ensures for authentication checks, API fetching, and assets preloading. The LinearProgressIndication provides essential feedback. The log is displayed as the first impression depicting recognition over recall to showcase app identity.

2.1.2 Login Page

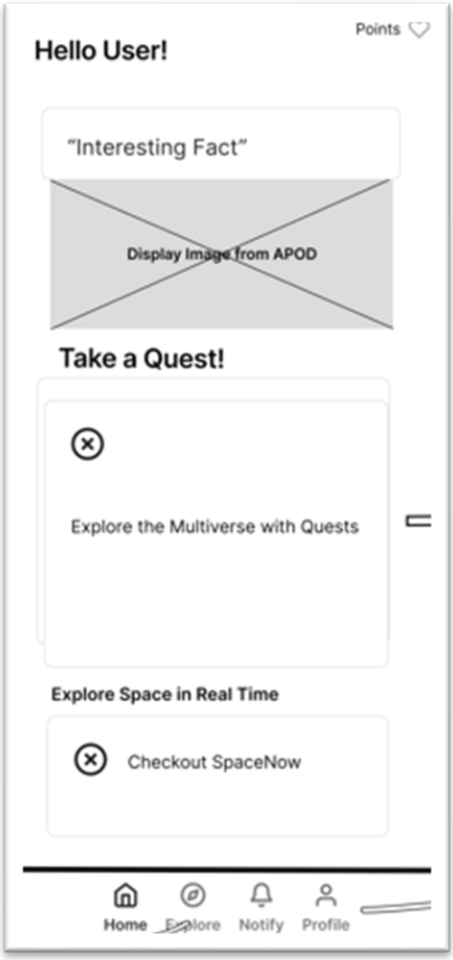
I have decided to implement authentication even though it's not a core requirement, a login page demonstrates understanding of real-world applications. The design emphasizes security, this design aligns with the usability of the app. Additionally, a forgot password option will be implemented if the user has forgotten their password.

Platform Differences:

This UI design is implemented as IOS, where text fields have placeholders, and in Android we use outlined Text fields. (Material Design, n.d.)

Error Validation will be implemented real time to validate users' inputs efficiently.

2.1.2 Home Screen



The primary functionality of the application is highlighted in the home page. Mainly the "Quest" This depicts the Visibility principle of HCI, ability of the user to understand and take appropriate actions easily. (dovetail.com, 2023) more stats and rank methods will be displayed in the Quest widget, may be developed in the future.

The Home Screen eliminates the need for a top navigation bar or a back button, since it represents the primary interface state, User can use bottom Navbar for accessibility across the application.

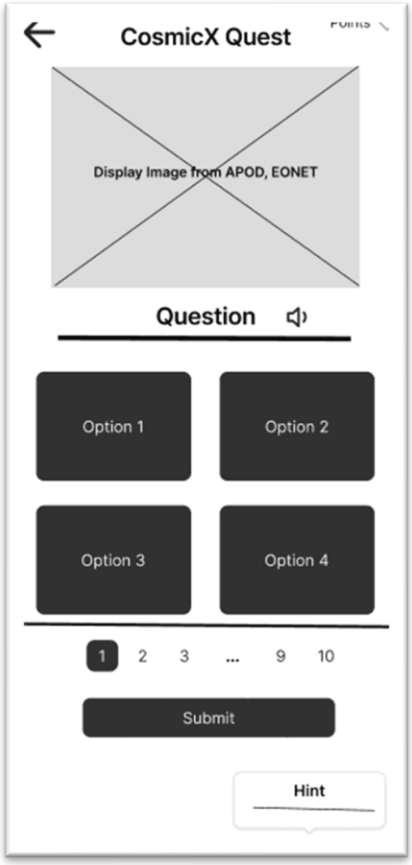
In the Bottom navbar consist platform differences. This visualization shows Android's bottom navbar. In IOS the bottom navbar mainly consist of SF icons and scalable tab bar icons (Apple Developer Documentation, n.d.)

As shown here the theme and the font specifications will always remain consist throughout the app. Here IOS use San Francisco but Android use Roboto Font. Reflecting the Consistency in HCI.

Image Source: Fetching NASA's APOD API to display the image when home screen is mounted.

Furthermore, the "Interesting Fact" text displays API's description field of the image

2.1.3 Quest Screen



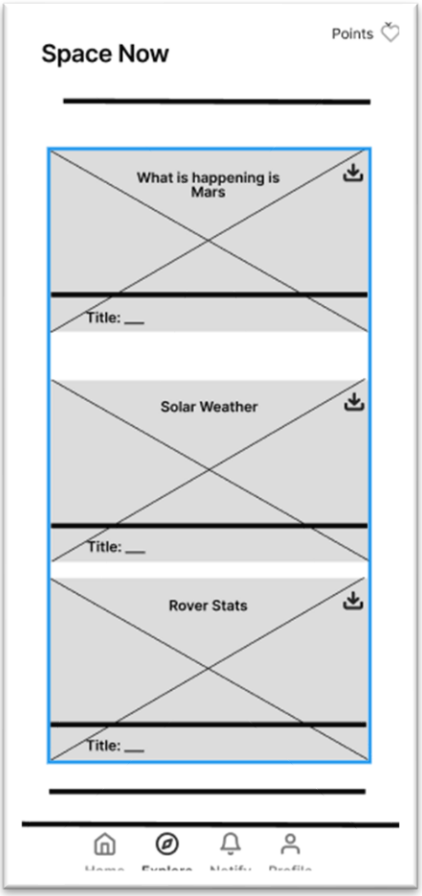
The Quiz Screen is the main engagement of the application. Design here mainly focuses on making the experience interactive and enjoyable for the users.

Image and Question: Questions will relate to the information fetched by the APIs and some questions will be implemented with a related Image. If the Image shows Mars, the question might ask about the stats on mars rover missions.

There are four widgets displaying the answers each widget is large in size to optimize the user experience.

Additionally, **a hint** will be shown for the ease of the user to guess the answer. This hint is extracted from the API's description that is used for the question.

2.1.4 Space Now explore



The Space Now screen provides a separate exploration platform. While quizzes deliver structured learning, Space Now screen enables self-exploration for more user interactivity.

Multi Source Content: Images and meta data are retrieved from multiple NASA APIs. Each widget will contain a image and metadata (title, data description, etc) which will be in the widget. Separate screen for each widget will be implemented if needed.

Future Enhancements: Data visualization using charts and interactive data such as live ISS tracker, Space weather and planetary positions.

2.1.5 Points

This Symbolizes the Points collected by doing Quests Points are displayed and maintained as global state in the application level in almost every interface. This will be incremented and shown when user completes Quests. will be used to calculate high scores and ranks. This will act as an intrinsic motivator, improving the engagement of users to improve their scores in the app. Ranking and badges will be awarded according to the points of the user

3. HCI Principles Justification

3.1 Usability

Each UI was designed and structured to reduce complexity and cognitive load to encourage intuitive exploration for kids and young people.

Figure 3.1.0

In the figure 3.1.0 is the loading screen here The Large SignIn button follows the Fitts Law large tap targets reduce input errors for kids, this is consistent in the whole application. (Experience, n.d.).

Figure 3.1.1

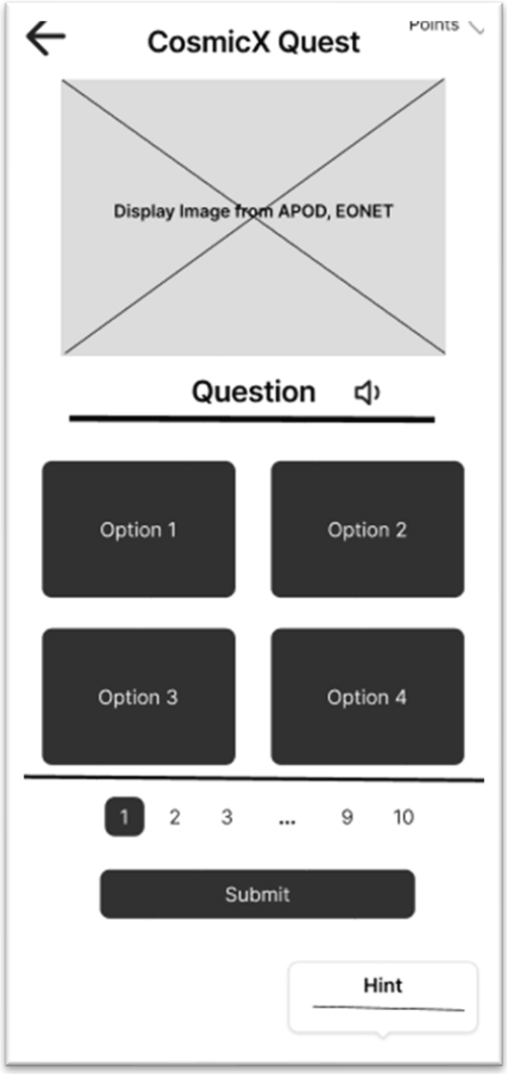


Figure 3.1.2

These interfaces prioritize learnability through efficient design choices. In the figure 3.1.1 the home screen has a persistent bottom navigation maintains **Visibility of System Status**, allows users to navigate through the app easily, the design uses consistent typography. Figure 3.1.2 depicts the Quiz Screen considered as the main functionality of the app, large answer buttons and adequate spacing follow both Visibility and **Error prevention** (Nielsen, 1994), when the user clicks an answer, the widgets changes colours indicating right and wrong answers as Green and Red, this supports the **Feedback principle**. The design throughout the app uses a NASA inspired colour palette for aesthetic consistency and minimalism. User Freedom in Nielsen's is depicted through the back navigation and the flexibility to customize profile settings.

Overall, these UI designs showcase how usability and feedback principles create an engaging and intuitive learning user experience throughout the app.

3.2 Accessibility

This design has ensured accessibility to the users, by maintaining a consistent layout patterns and simple typography for quick recognition. Prioritizing equality among the users for interactivity and navigation specially for people with visual impairments and auditory disabilities. Below accessibility features aligns with the **WCAG 2.1 guidelines** which are crucial for accessibility standards. (Nielsen, 1994)

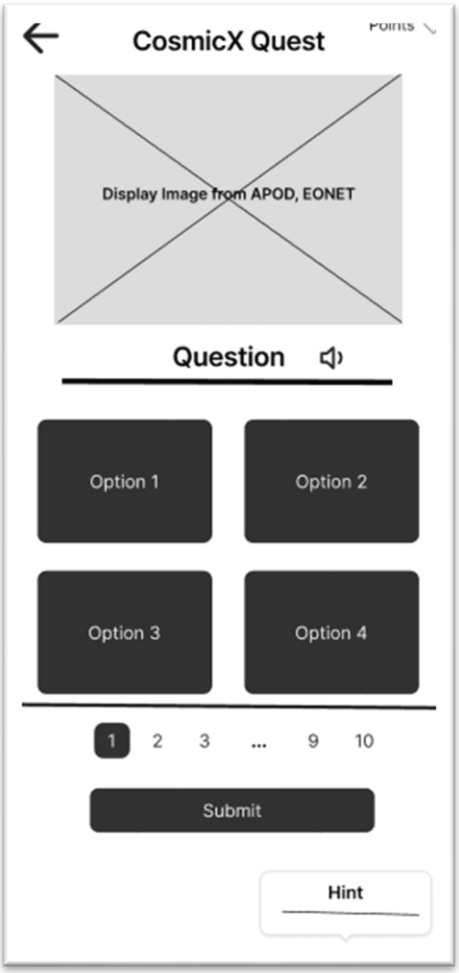


Figure 3.2.1

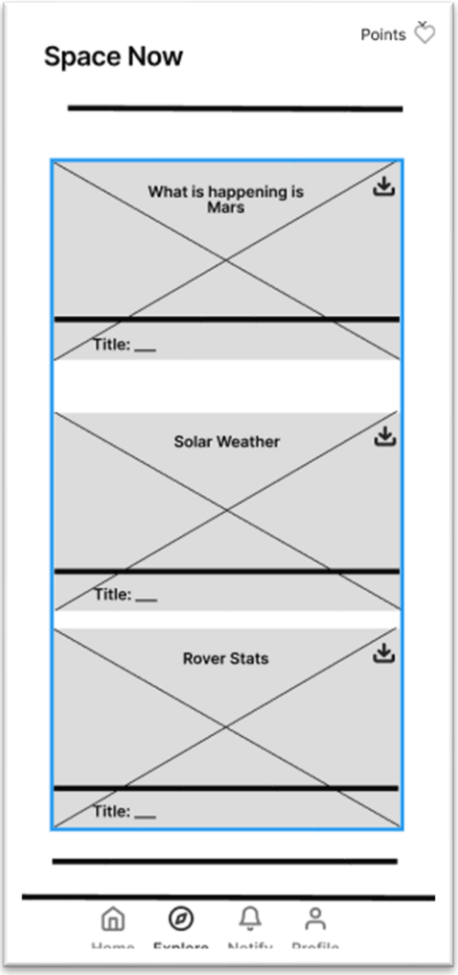


Figure 3.2.2

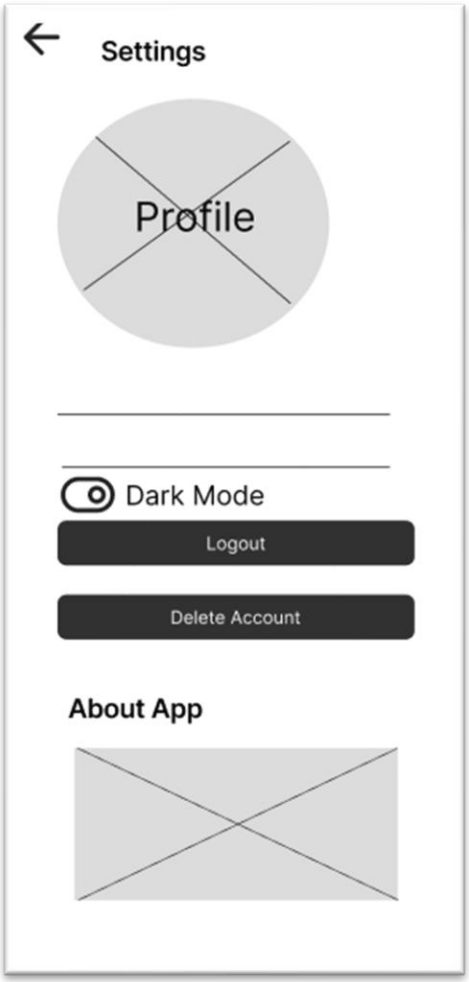


Figure 3.2.3

As shown in the figure 3.2.1 the Quiz screen, A sound button will be implemented as a Text to Speech feature. This enhances accessibility for visually impaired users and the large widget designs aligns with **Flexibility and Efficiency of Use**, making the app adaptable to the user's needs.

In the Explore page figure 3.2.2, Alt text description and offline image caching support users who have limited access to internet facilities. Emphasizing equality among the users.

One of the most valuable accessibility features in the modern-day applications is the dark mode feature. It is beneficial for users with visual impairments and for other purposes (Codelantic.com, 2020). In the figure 3.2.3 the Settings page consists of a toggle button for the Dark mode enabling feature this reduces eye strain and glare. It is also important since the target audience of the app are youngsters, the app has to be comfortable for them.

3.3 Cross Platform Consistency

The UI designs leverages Flutter's adaptive widget design system to maintain a native cross platform consistently between Android and IOS and adhering to their design guidelines. Considering both platforms, the main difference is that Android uses the Material 3 design patterns and IOS adopts to Cupertino component style, making users feels like home at their own respective platforms. All the other changes are mentioned in the annotation provided in the wire frame design. Despite these native changes, core visual elements will remain the same.

4. Conclusion

In conclusion, CosmicX design illustrates how the HCI principles can enhance the user's experience, engagement, interactivity and learning. Integrating cross platform UI components, accessible interactivity ways, and engagement through gamification. CosmicX design provides users an educational space exploration platform by efficiently justifying HCI principles and cross platform consistency.

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